

Multimodal Music Classification

AI in Music, Winter Term 2025/2026

27.11.2025

Exercise 1 [Genre Classification with Album Images, 4 points]

Download the dataset provided here:

<https://rwth-aachen.sciebo.de/s/fSbKAqRaTJypY7P>.

It contains 30 songs (in mp3 format) and their corresponding album covers (in webp format) from the genres Rock and Classical. These songs were taken from the Free Music Archive (FMA).

Use the scikit-image python package to extract the following features for each album cover:

- **HSV hue:** mean and variance
- **HSV saturation:** mean and variance
- **HSV value:** mean and variance
- **Edge density:** Use the Canny edge detection algorithm (`skimage.feature.canny`), sum how many pixels are detected as edges and then normalise that value by the total amount of pixels

Then, do the following on 300 different random seeds:

Create train and test sets with 20 songs for training and 10 songs for testing. Ensure that the test set remains balanced (5 songs from each genre) by using a stratified split. Train a Random Forest (RF) classifier with the sklearn implementation of `RandomForestClassifier` to detect which genre the images belong to. You can use the default parameter values for all sklearn and skimage functions you need.

Evaluate your models by calculating the mean accuracy and F1-scores across the 300 classifiers you trained.

Exercise 2 [Genre Classification with MFCCs, 4 points]

Use librosa to calculate the Mel-frequency Cepstral Coefficients (MFCCs) of the songs. Then, calculate the mean and variance across the first four rows of the MFCC matrix. Use the resulting eight features to train 300 RF genre classifiers again, as in Exercise 1, and evaluate mean accuracy and F1-scores.

Exercise 3 [Genre Classification with Album Images and MFCCs, 2 points]

In a final experiment, stack the image and audio features into a single feature vector for every song. Again, train 300 RF genre classifiers with these combined feature vectors and calculate mean accuracy and F1-scores.

Does the use of multimodal data improve the classification performance of your models in this task?